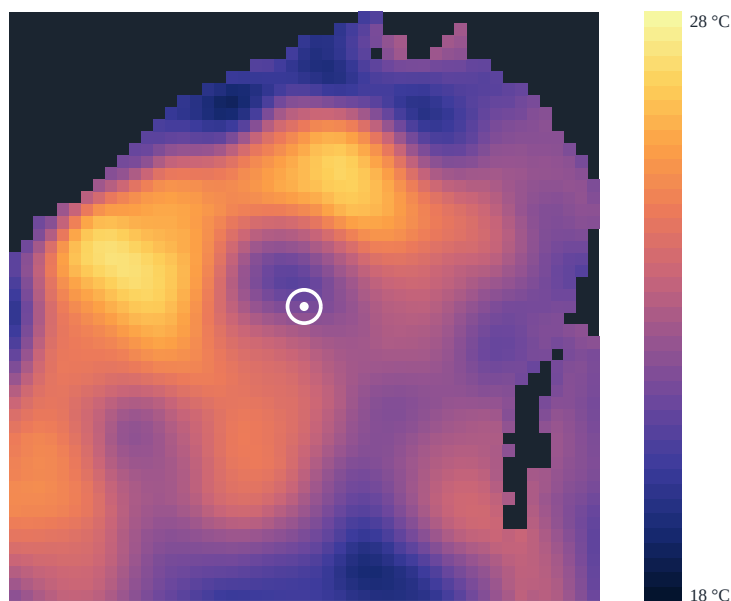


Water-column report

Location	15.00°N, 88.00°E (grid cell 15.125°N, 88.125°E)
Date	2023-05-11
Map depth	100 m
Variables	temperature (reconstructed) · ± 1 sd uncertainty (estimated) · climatology (baseline) · MLD, D20, D26, TCHP (derived)
Observations	Argo 2902766 (measured) · 82.0 km away · +3 d · held-out year (independent)
Model	cnn-unet-v1 (U-Net reconstruction) · data 2019-01-01..2023-12-31

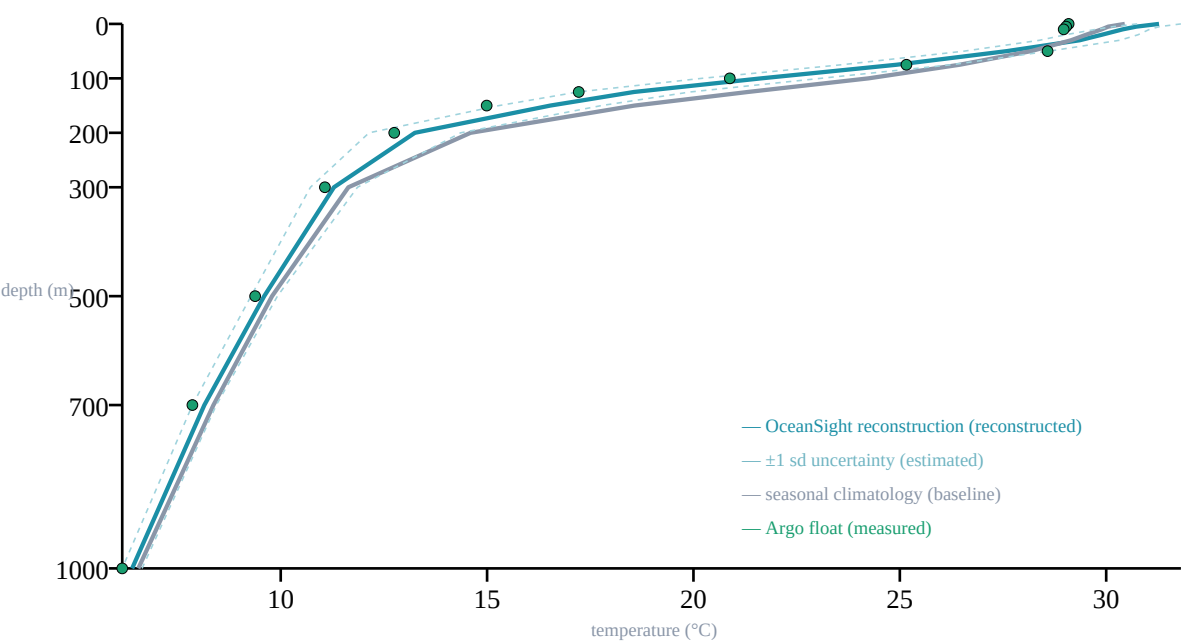
Reconstructed surface temperature is 31.3°C with a mixed layer to about 20 m. The 26°C isotherm sits near 65 m, the 20°C isotherm (thermocline proxy) near 113 m, giving moderate cyclone heat potential (77 kJ/cm²). The surface is 0.8°C warmer than the pre-monsoon seasonal climatology for this date.

Location on the reconstructed field



82.1°E 94.1°E
Reconstructed temperature at 100 m · 9.1–21.1°N · white ring = report location

Temperature profile



Derived structure · derived from the reconstructed column

TCHP	MLD	D26	D20
76.6 kJ/cm²	20.0 m	64.9 m	113.3 m

Values at the standard depths

depth (m)	T (°C) · reconstructed	±1 sd (°C) · estimated	climatology (°C)	Argo (°C) · measured
0	31.28	0.53	30.45	29.09
5	30.73	0.56	30.05	29.03
10	30.40	0.65	29.89	28.97
20	29.89	0.88	29.51	—
30	29.36	0.97	29.15	—
50	27.59	1.03	28.18	28.58
75	24.92	1.33	26.45	25.16
100	21.64	1.44	24.24	20.88
125	18.56	1.36	21.32	17.22
150	16.53	1.24	18.57	14.99
200	13.25	1.11	14.60	12.75
300	11.29	0.57	11.64	11.07
500	9.60	0.32	9.79	9.38
700	8.15	0.29	8.37	7.86
1000	6.40	0.24	6.55	6.16

Method, provenance and validation

Temperature at 15 standard depths (0–1000 m) is reconstructed on a 0.25° daily grid from satellite surface observations (sea-surface temperature, salinity, sea level, geostrophic currents, winds) by a U-Net trained on the HYCOM ocean analysis (2019–2021). The model predicts the departure from a seasonal climatology and a per-depth uncertainty (one standard deviation, sd), calibrated against 2022 Argo profiles. Derived quantities: MLD — first depth below 10 m that is 0.5 °C colder than 10 m; D20 / D26 — depth of the 20 / 26 °C isotherm; TCHP — heat content above 26 °C.

Measured — direct observation (Argo floats). **Reconstructed** — OceanSight model output (temperature, uncertainty). **Derived** — calculated from the reconstructed column (MLD, D20, D26, TCHP). **Estimated** — inferred quantity with additional uncertainty (calibrated ±1 sd). Independent Argo validation (2023-01-01..2023-12-31, 2639 held-out profiles): RMSE 1.31 °C at 100 m. Validation floats are independent of model training but not fully independent of the ocean analysis used as the training target (it assimilates Argo). OceanSight reconstructs past ocean states; it is not a forecast.

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